

Mediroza Hospital Week 4 Report

Penetration Testing Project | Batch B083

Target	https://medirozahospital.com
Prepared by	Vindhya
Report date	2 October 2026
Classification	Confidential training evidence
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Organisation	Networkwalks
Batch	B083 Week 4 Capstone Project

This report is produced as part of a controlled educational exercise. The target was authorised for security testing. These techniques must never be applied to any system without explicit written permission from the owner.

01. Executive Summary

I conducted a black-box penetration test against the web infrastructure of Mediroza General Hospital at <https://medirozahospital.com>, as authorised under the Week 4 capstone project brief. The objective was to identify vulnerabilities, demonstrate their real-world impact through controlled exploitation, and provide actionable recommendations.

During the assessment I identified an authentication bypass vulnerability on the patient portal login page, which allowed access to the portal without valid credentials. This exposed three confidential patient lab report PDFs, each protected by a weak password. After recovering the passwords, metadata embedded in one of the files revealed an internal reference to a backup location on the server. Following that lead, I located a publicly accessible folder containing a full database backup, which exposed staff salary records and shareholder ownership details in plain text.

This confidential instructor submission includes the supplied screenshots without redaction, as requested. The evidence appendix preserves the original images, including the exercise passwords and the patient, staff and shareholder details visible in them.

Overall Risk Rating: CRITICAL — immediate remediation recommended.

02. Scope and Methodology

Scope

Testing was limited to the authorised target domain: <https://medirozahospital.com>. Social engineering, denial-of-service attacks, and any testing outside the agreed domain were explicitly excluded, per the project brief.

Methodology

- Reconnaissance and follow-up — reviewed robots.txt and the /old/ HTTP response while investigating the server backup location.
- Vulnerability Identification — analysing authentication behaviour and input handling on the patient portal login page.
- Exploitation — demonstrating real impact via a login bypass, followed by password recovery on the retrieved files and analysis of file metadata.
- Documentation — recording all findings, evidence, and remediation recommendations in this report.

Tools Used

- curl / browser — for reconnaissance and inspecting HTTP responses (robots.txt, directory listing)
- Patient portal login form — for testing authentication behaviour and injection
- Networkwalks Hash Calculator and Password Cracker — extracted PDF hashes and recovered passwords through dictionary attacks in the browser.
- WPS PDF viewer — displayed opened reports. qpdf was attempted in Kali but was unavailable; the screenshot does not show successful qpdf decryption.
- exiftool — for reading hidden metadata in PDF files
- wget — for downloading the exposed backup file
- file / sha256sum / grep / less / sed — validated the backup, located tables, viewed records, and attempted text exports.
- AI-assisted reporting — organized the supplied evidence and drafted explanations and recommendations.

Limitations

The evidence includes SQL error responses, portal access, browser-based password recovery, PDF metadata, backup download and SQL records. It does not include a captured HTTP login request or backend query, an opened page from PDF 3, the entire employee salary list, or validation of the exported text files. All 21 images are retained, including troubleshooting attempts, with captions distinguishing successful results from errors.

03. Findings and Proof of Exploitation

Summary Table

#	Vulnerability	Location	Risk	Status
1	Authentication bypass on patient login	patient/login.php	Critical	Confirmed
2	Confidential PDFs accessible after bypass	patient portal	High	Confirmed
3	Weak PDF passwords recoverable via wordlist attack	patient_report_*.pdf	High	Confirmed
4	Sensitive metadata left in patient PDF	patient_report_3.pdf	Medium	Confirmed
5	Publicly accessible backup folder	/old/	Critical	Confirmed
6	Staff salary & shareholder data exposed in backup	/old/*.sql	Critical	Confirmed

Finding 1 — Authentication Bypass on Patient Login

Risk Rating: Critical | Location: patient/login.php

I tested the patient login with a quote in the username. E01a shows admin ' alongside a mysqli_query() SQL syntax warning. E01b shows a revised username containing quote and hyphen characters while the warning remains visible. These responses indicate SQL input-handling and error-disclosure weaknesses. E01 shows the subsequent portal access after the reported bypass, and E02 shows the report download list. Together, the screenshots support the SQL injection investigation and access outcome; they do not expose the underlying backend query.

Evidence sequence: E01a shows the initial quote input and SQL warning; E01b shows the revised login input; E01 shows the subsequent portal access. The warning visible in E01b is not itself proof of a successful login.

See E01a, E01b and E01 in the evidence appendix for the login investigation and portal access result.

Finding 2 — Confidential PDFs Accessible After Bypass

Risk Rating: High | Location: patient portal

After bypassing authentication, three confidential patient lab report PDFs were visible and downloadable from the portal. These are medical documents that should only be accessible to the named patients and their doctors.

See E02 for all three report links in the patient portal.

Finding 3 — Weak PDF Passwords Recoverable via Wordlist Attack

Risk Rating: High | Location: patient_report_*.pdf

I used the Networkwalks Hash Calculator to extract PDF hashes, then the Networkwalks Password Cracker to run dictionary attacks. Reports 1 and 2 used the built-in 100-entry list, recovering 123456 and password respectively. For report 3, I used rockyou.txt; E09a shows its download page. The recovery result in E09 shows a successful match for !@#\$\$%^& after 72,203 candidates from a list of 14,344,379 entries. These are recovered weak passwords, not a break of the encryption algorithm itself.

See E03–E09 and E09a for hash extraction, wordlist selection, password recovery and opened report evidence. The images show opened pages for reports 1 and 2; report 3 is supported by its password match and successful metadata read in E12.

Finding 4 — Sensitive Metadata in Patient PDF

Risk Rating: Medium | Location: patient_report_3.pdf

ExifTool initially warned that patient_report_3.pdf was password protected (E11). Supplying the recovered password with the -password option allowed the metadata to be read directly from the encrypted PDF (E12). Its Comments field stated: DB backup moved to /old before site migration, do not delete. The preceding qpdf attempt failed because the tool was not installed (E10).

See E11–E12 for the initial warning and successful password-assisted metadata output.

Finding 5 — Publicly Accessible Backup Folder

Risk Rating: Critical | Location: /old/

The /old/ path was listed in robots.txt, which is a crawler instruction rather than an access control. Requesting /old/ returned HTTP 200 and an Index of /old/ listing that linked to mediroza_db_backup_2019.sql. A curl

download transferred 6,346 bytes. Subsequent successful file and sha256sum commands identified ASCII text and produced an integrity hash. See E10 and E13–E15.

E10 — robots.txt lists /patient/, /staff/ and /old/. The same image records a failed qpdf installation attempt.

E13 — HTTP 200 directory-index response exposes the SQL backup filename and link.

E14 — completed curl download and successful file-type and SHA-256 checks after earlier command errors.

Finding 6 — Staff Salary and Shareholder Data Exposed in Backup

Risk Rating: Critical | Location: /old/mediroza_db_backup_2019.sql

The backup exposed staff and shareholders tables. E15 locates their definitions at lines 15 and 69. E16 shows staff fields, including monthly_salary_zar, and part of the employee records. E17 shows shareholder names, percentages, share counts and share classes. The participant reports completing both assigned reviews; the attached staff view is only a partial extract, so it does not demonstrate every employee salary. E18 records attempts to export both INSERT sections to text files, but does not display or verify the resulting file contents.

See E15–E17 for table discovery and the displayed staff and shareholder records.

See E18 for the text-export commands. The search patterns omit the SQL identifier backticks visible in E16–E17, so the resulting files may be empty; a returned prompt alone does not establish a successful extraction.

04. Risk Rating

These qualitative ratings reflect the sensitivity of the exposed data and the access demonstrated in the exercise; they are not calculated CVSS scores. Authentication bypass is rated Critical on the reported access path, with the evidence limitation stated in Finding 1. Directory listing and backup data exposure are connected findings within the same exposure chain.

#	Vulnerability	Location	Risk	Status
1	Authentication bypass	—	Critical	Unauthenticated full access to patient portal with trivial payload
2	Exposed patient PDFs	—	High	Confidential medical records reachable with no valid credentials
3	Weak PDF passwords	—	High	Passwords recovered through the browser dictionary tool; exact elapsed times were not captured.
4	Metadata leakage	—	Medium	Internal note disclosed a sensitive server path, no direct access gained
5	Public backup folder	—	Critical	Directory listing exposed a full database backup to anyone
6	Salary/shareholder data in backup	—	Critical	Highly sensitive financial and personal data unprotected in plain text

05. Recommendations and Remediation

Fix Authentication Bypass

Use parameterised queries or prepared statements so login input cannot alter SQL query structure. Return a generic login error to the browser and log technical database errors securely on the server.

Fix PDF Access and Password Strength

Serve patient reports through authenticated access checks and enforce authorization for the specific patient and document. Store the files outside the public web root where practical. Use long, unique, unpredictable document passwords when password protection is required, rather than common words or symbol sequences.

Strip PDF Metadata

Remove internal comments and unnecessary metadata before distribution. Regenerate or sanitize the final PDF and inspect the resulting file to verify that sensitive metadata cannot be recovered.

Fix Directory Listing and Remove the Backup

Disable directory listing on all web server folders (e.g. Options -Indexes in Apache configuration). Delete the exposed backup immediately. Database backups must never be stored inside the web root — store them in a private, access-controlled location.

Protect Sensitive Internal Data

Encrypt database backups containing personal and financial data at rest. Apply the principle of least privilege so that salary and shareholder records are accessible only to authorised personnel, and audit for other forgotten files in similarly disallowed paths.

This report is produced as part of a controlled educational exercise. The target has been authorised for security testing. These techniques must never be applied to any system without explicit written permission from the owner.

E01a M1 Initial access

Initial quote input produces a SQL error

The username field contains admin ' and the page displays a mysqli_query() SQL syntax warning. This error guided further testing of the login input.

Patient Portal

Sign in to view your lab reports and results.

Warning: mysqli_query(): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''' at line 1

Username

Password

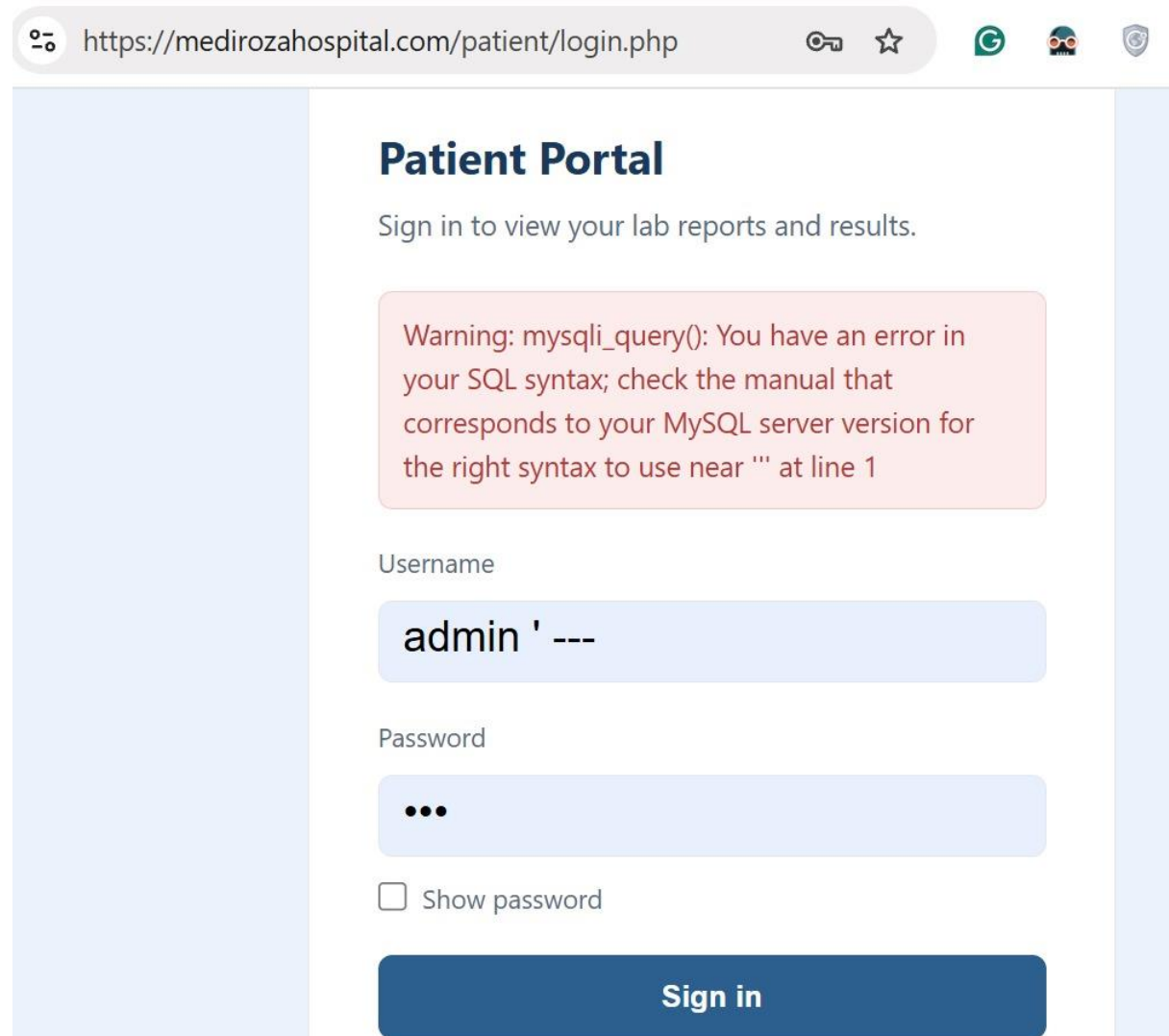
☐ Show password

Sign in

E01b M1 Initial access

Revised login input during SQL injection testing

The login page shows a revised username with quote and hyphen characters while the SQL warning is still displayed. Successful portal access is shown separately in E01.



https://medirozahospital.com/patient/login.php

Patient Portal

Sign in to view your lab reports and results.

Warning: mysqli_query(): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''' at line 1

Username

admin ' ---

Password

...

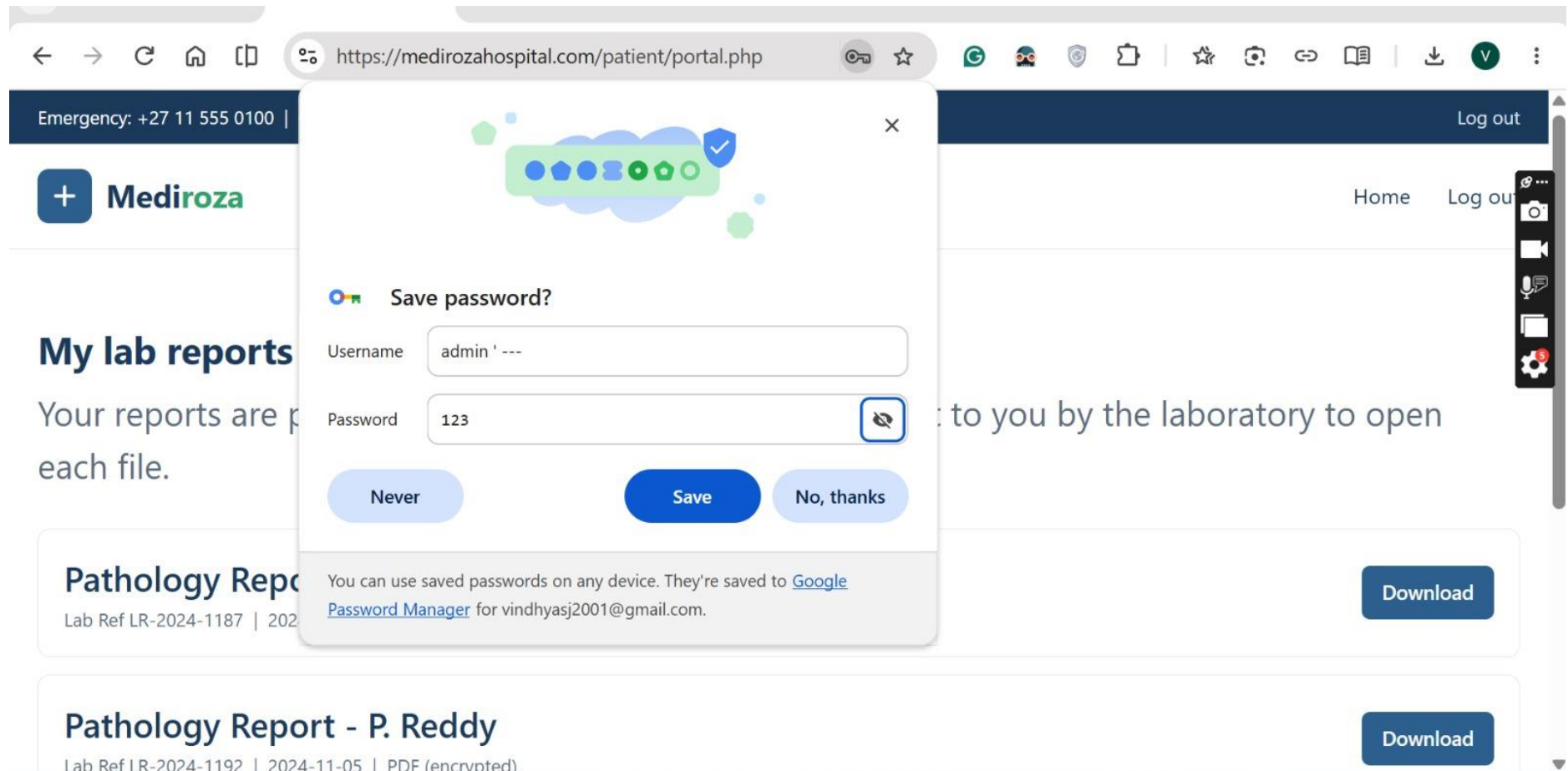
☐ Show password

Sign in

E01 M1 Initial access

Patient portal after the reported login bypass

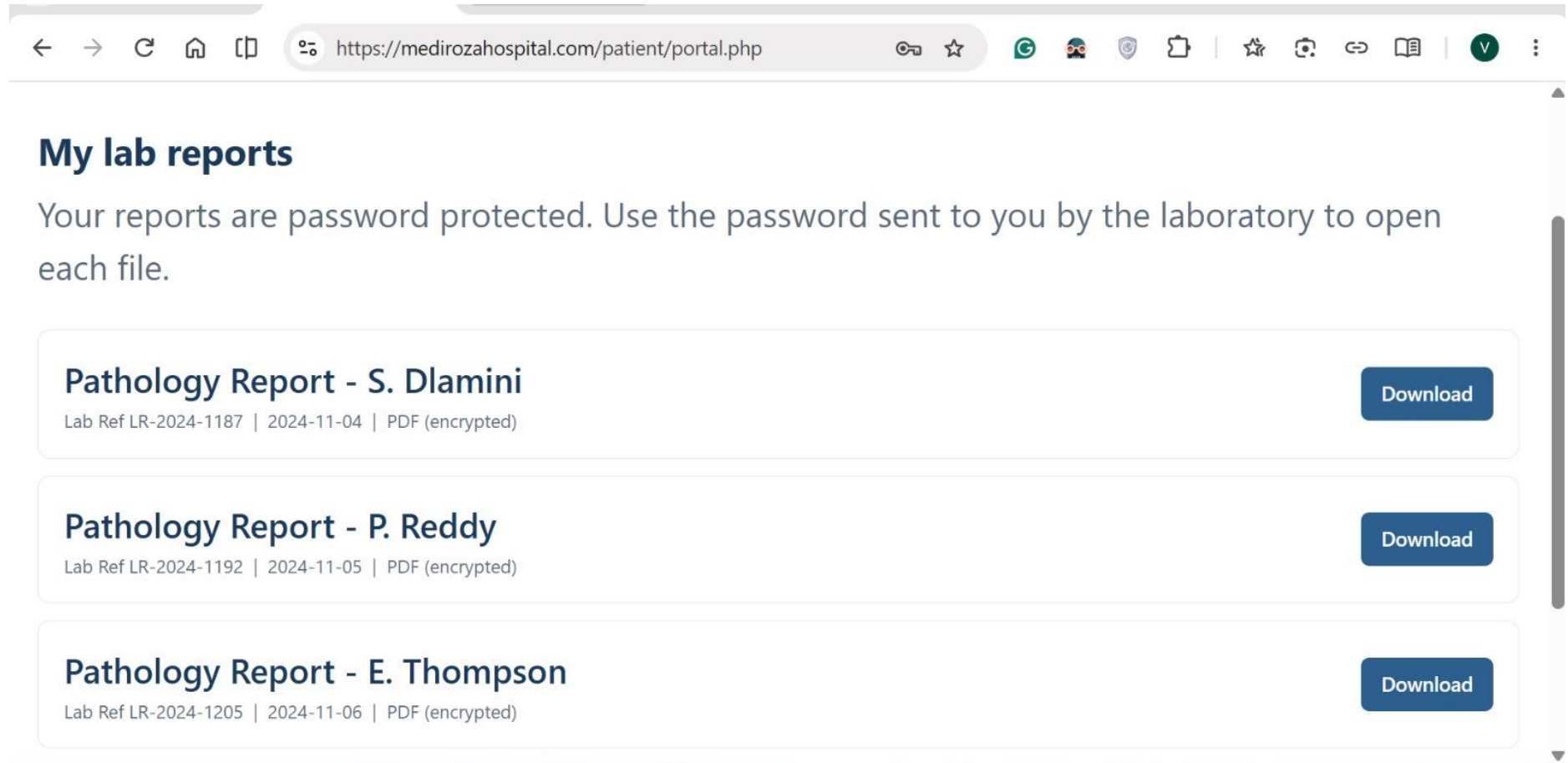
The browser save-password prompt shows the crafted username while the patient portal is open. This records the access outcome following the login investigation shown in E01a and E01b.



E02 M1 Initial access

Three confidential PDF report download links

The portal lists three encrypted pathology reports with Download buttons, establishing the files exposed after access.



The screenshot shows a web browser window with the address bar displaying <https://medirozahospital.com/patient/portal.php>. The page title is "My lab reports". Below the title, a message states: "Your reports are password protected. Use the password sent to you by the laboratory to open each file." There are three report cards listed, each with a title, a subtitle, and a "Download" button.

Report Title	Lab Ref	Date	Format	Action
Pathology Report - S. Dlamini	LR-2024-1187	2024-11-04	PDF (encrypted)	Download
Pathology Report - P. Reddy	LR-2024-1192	2024-11-05	PDF (encrypted)	Download
Pathology Report - E. Thompson	LR-2024-1205	2024-11-06	PDF (encrypted)	Download

E03 M2 PDF password recovery

Report 1 hash extraction

The Networkwalks Hash Calculator identifies patient_report_1.pdf as encrypted and displays a crackable PDF hash.

← → ↻ 🏠 📄 🔍 https://networkwalks.com/hash-calculator/ 🔍 ☆ 🟢 🧑 🛡️ 📁 | ☆ 📷 🔗 📖 | ⬇️

Hash Calculator

 Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF 

TextFilePDF



Click to choose a PDF, or drop one here
Nothing is uploaded, the PDF is parsed locally

patient_report_1.pdf | 3.6 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

\$pdf\$2*3*128*4294967292*1*32*3361663365326235643333353531613238303137316238333238373763353339*32*ef16c52ab8efce2c18c79e9d28895b5928bf4e5e4e758a4164004e56fffa0108*32*c431fab9cc5ef7b59c244b61b745f71ac5ba427b1b9102da468e77127f1e69d6

CopyDownload

Revision: R3 Version: V2 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

E04 M2 PDF password recovery

Report 1 password recovered

The Networkwalks Password Cracker shows a match for 123456 using its built-in 100-entry wordlist.

The screenshot shows the Networkwalks Password Cracker web application in a browser. The URL bar displays <https://networkwalks.com/password-cracker/>. The application interface has a dark theme with a background of binary code. At the top, the title "Password Cracker through Dictionary Attacks" is displayed, along with a brief description: "Hash every word in a wordlist and match it against a PDF password hash, the same idea John the Ripper uses." Below the title, there are three numbered steps: 1. Extract the \$pdf\$ hash in the Hash Calculator, 2. Paste it below, and 3. Run the attack, then open your PDF with the password. The main area shows the progress of the attack: "Tried: 1 / 100" and "200 pw/s". A log of the attack process is displayed, showing the loaded wordlist, target, and the successful match: "[+] MATCH 123456 ✓". Below the log, a large green box displays the password "123456" with the text "PASSWORD CRACKED SUCCESSFULLY" above it. A message below the password states: "This is the PDF password. Open your PDF and enter it to unlock the file." A "Copy password" button is located below the message. At the bottom, the PDF hash is shown: "\$pdf\$2*3*128*4294967292*1*32*3361663365326235643333353531613238303137316238333238373763353339*32*e*f16c52ab8efce2c18c79e9d28895b5928bf4e5e4e758a41648004e56ffffa0108*32*c431fab9cc5ef7b59c244b61b745f71ac5ba427b1b9182da468e77127f1e69d6". At the bottom of the interface, there are buttons for "Built-in list (100 passwords)" and "Upload wordlist (.txt)", and a status bar indicating "Active: built-in list, 100 words".

networkwalks
Dictionary Attack LAB
Password Cracker through Dictionary Attacks
Hash every word in a wordlist and match it against a PDF password hash, the same idea John the Ripper uses.

1 Extract the \$pdf\$ hash in the Hash Calculator 2 Paste it below
3 Run the attack, then open your PDF with the password

Tried: 1 / 100 200 pw/s 1%

[*] Loaded wordlist: built-in list (100 entries)
[*] Target: PDF R3 / 128-bit / dictionary attack
[*] Deriving keys with MD5 + RC4 ...
[+] MATCH 123456 ✓

PASSWORD CRACKED SUCCESSFULLY

123456

This is the PDF password. Open your PDF and enter it to unlock the file.

Copy password

PDF HASH (\$PDF\$...)

\$pdf\$2*3*128*4294967292*1*32*3361663365326235643333353531613238303137316238333238373763353339*32*e*f16c52ab8efce2c18c79e9d28895b5928bf4e5e4e758a41648004e56ffffa0108*32*c431fab9cc5ef7b59c244b61b745f71ac5ba427b1b9182da468e77127f1e69d6

Built-in list (100 passwords) Upload wordlist (.txt) Active: built-in list, 100 words

E05 M2 PDF password recovery

Report 1 opened in the PDF viewer

The readable pathology report supports successful access to its contents. Evidence is grouped by workflow rather than capture timestamp.

WPS PDF Viewer Interface:

- Top Bar: WPS logo, Exclusive Massive Perks, PDF icons, Upgrade Now button.
- Menu: File, Home, Edit, Page, Comment, Tool, Fill & Sign, Protect, Convert, WPS AI, Search, Not uploaded, Share.
- Home Tab: Edit Content, Add Text, Table recognition, PDF Converter, Highlight, Text Comment, Sign, OCR PDF, Extract Text, AI Proofing, View.
- Left Panel: Thumbnail view showing a preview of the report.
- Right Panel: Search, Zoom, and other navigation tools.

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PATHOLOGY LABORATORY REPORT CONFIDENTIAL

Patient Name: **Sipho Dlamini** Report Date: **2024-11-04**
Patient ID: **MG-P-10231** Referring Dr: **Dr. Anita Naicker**
Date of Birth: **1984-06-12** Specimen: **Whole blood (EDTA)**
Gender: **Male** Lab Ref No: **LR-2024-1187**

TEST	RESULT	UNIT	REFERENCE RANGE	FLAG
Haemoglobin	13.9	g/dL	13.0 - 17.0	Normal
White Cell Count	11.8	$\times 10^9/L$	4.0 - 11.0	HIGH
Platelet Count	245	$\times 10^9/L$	150 - 400	Normal
Fasting Glucose	5.4	mmol/L	3.9 - 5.5	Normal
Creatinine	88	umol/L	64 - 104	Normal

Page 1/1, 74% zoom.

E06 M2 PDF password recovery

Report 2 hash extraction

The Networkwalks Hash Calculator displays the extracted hash for patient_report_2.pdf and its 128-bit key-length information.

Hash Calculator



Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



TextFilePDF



Click to choose a PDF, or drop one here
Nothing is uploaded, the PDF is parsed locally

patient_report_2.pdf | 3.5 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

```
$pdf$2*3*128*4294967292*1*32*31663463383732363564376264643638346637373032  
65633666363264616463*32*39f4e6b0aedef12344c340ffb39f8905528bf4e5e4e758a416  
4004e56fffa0108*32*408b37bcf12da873d7f2840f3c1b917a023961ded4c8164d38e46e  
9655e66775
```

CopyDownload

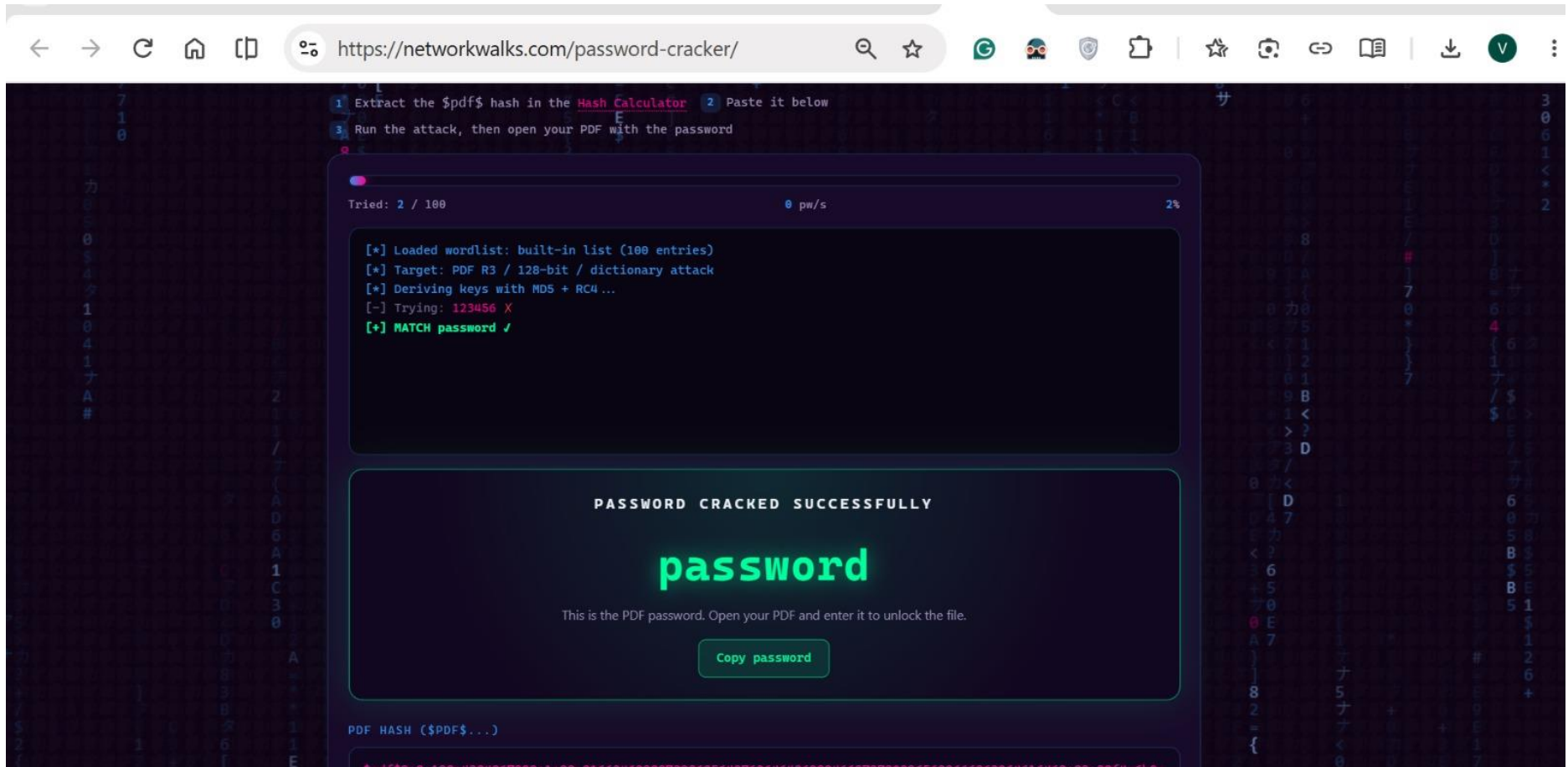
Revision: R3 Version: V2 Key length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash

E07 M2 PDF password recovery

Report 2 password recovered

The browser dictionary tool reports a match for password after trying two entries from the built-in list.



E08 M2 PDF password recovery

Report 2 opened in the PDF viewer

The second pathology report is readable. This demonstrates document access without adding any medical interpretation.

The screenshot shows a WPS AI PDF viewer interface. The top toolbar includes options like Home, Edit, Page, Comment, Tool, Fill & Sign, Protect, Convert, WPS AI, and a search icon. Below the toolbar, there are more specific tools like Edit Content, Add Text, Table recognition, PDF Converter, Highlight, Text Comment, Text Box, Sign, OCR PDF, Extract Text, AI Proofing, and View. The main content area displays a pathology report from Mediroza General Hospital. The report is titled 'PATHOLOGY LABORATORY REPORT' and is marked 'CONFIDENTIAL'. It contains patient information, test results, and a table of cholesterol and glucose levels.

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PATHOLOGY LABORATORY REPORT CONFIDENTIAL

Patient Name: **Priya Reddy** Report Date: **2024-11-05**
Patient ID: **MG-P-10244** Referring Dr: **Dr. Johan van der Merwe**
Date of Birth: **1991-02-28** Specimen: **Serum**
Gender: **Female** Lab Ref No: **LR-2024-1192**

TEST	RESULT	UNIT	REFERENCE RANGE	FLAG
Total Cholesterol	6.4	mmol/L	< 5.0	HIGH
LDL Cholesterol	4.1	mmol/L	< 3.0	HIGH
HDL Cholesterol	1.2	mmol/L	> 1.0	Normal
Triglycerides	1.8	mmol/L	< 1.7	HIGH
Fasting Glucose	4.9	mmol/L	3.9 - 5.5	Normal

E09a M2 PDF password recovery

rockyou.txt wordlist used for report 3

I used rockyou.txt for the third PDF dictionary attack. This screenshot shows its Weakpass download page, including the compressed download and checksums. The password recovery result follows in E09.

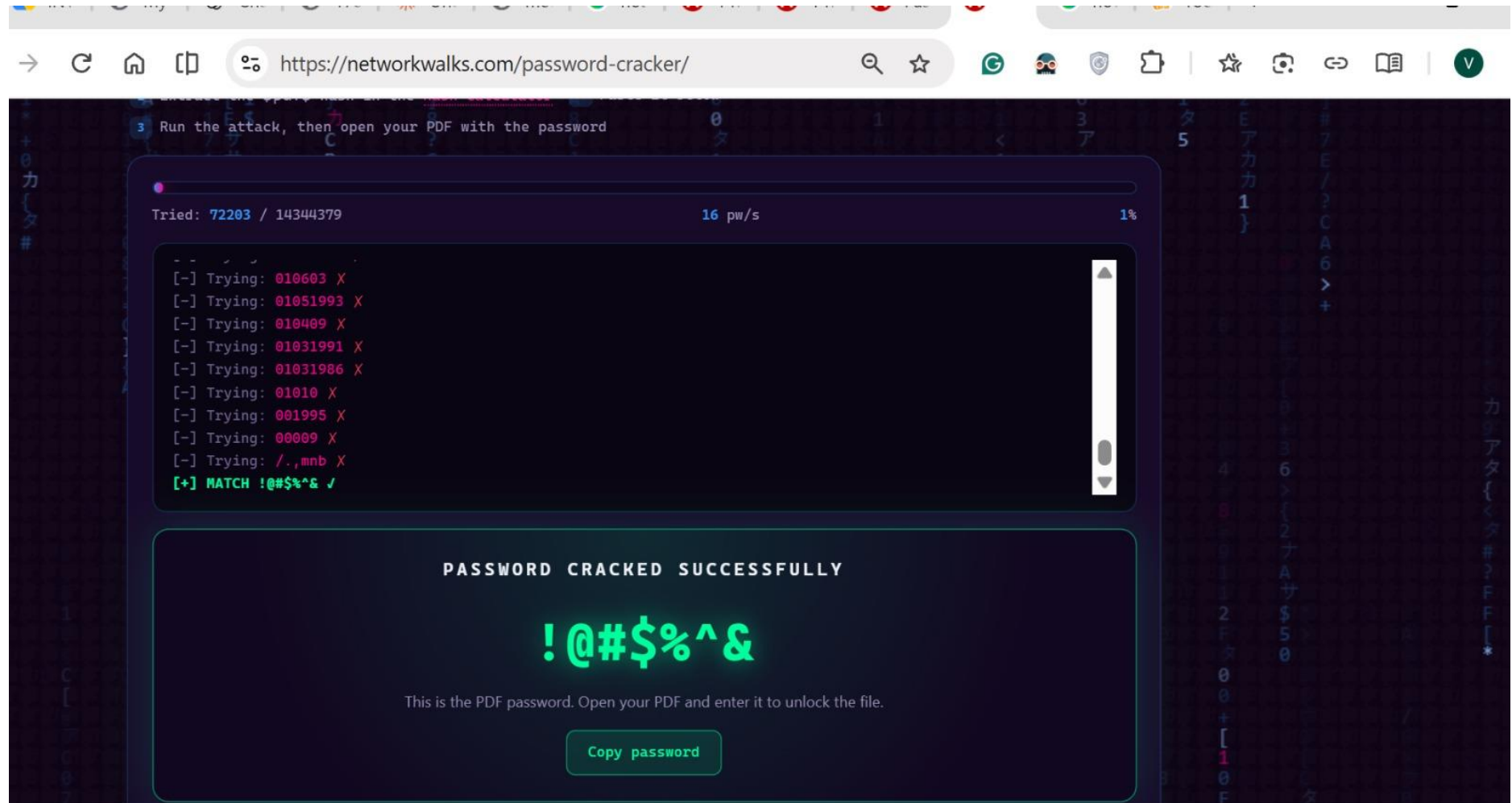
The screenshot shows the Weakpass website interface for downloading the rockyou.txt wordlist. The browser address bar displays `https://weakpass.com/wordlists/rockyou.txt`. On the left, a sidebar contains navigation links: `./wordlist`, `./download`, `./checksums`, `./length`, `./masks`, `./top-masks`, and `./examples`. The main content area features large statistics at the top: `251987`, `14.34M`, `133.44 MB`, and `50.89 MB`. Below these, the `$ download` section offers two options: `[ rockyou.txt.gz ]` and `rockyou.txt.gz.torrent`. The `$ checksums` section contains a table with MD5 hashes for both compressed and uncompressed versions. The `$ password length` section is partially visible at the bottom.

File	Checksum Type	Checksum	Action
rockyou.txt.gz	MD5	a5a18cce9faec1639e1dea86dcac1f95	copy
rockyou.txt	MD5	9076652d8ae75ce713e23ab09e10d9ee	copy

E09 M2 PDF password recovery

Report 3 password recovered

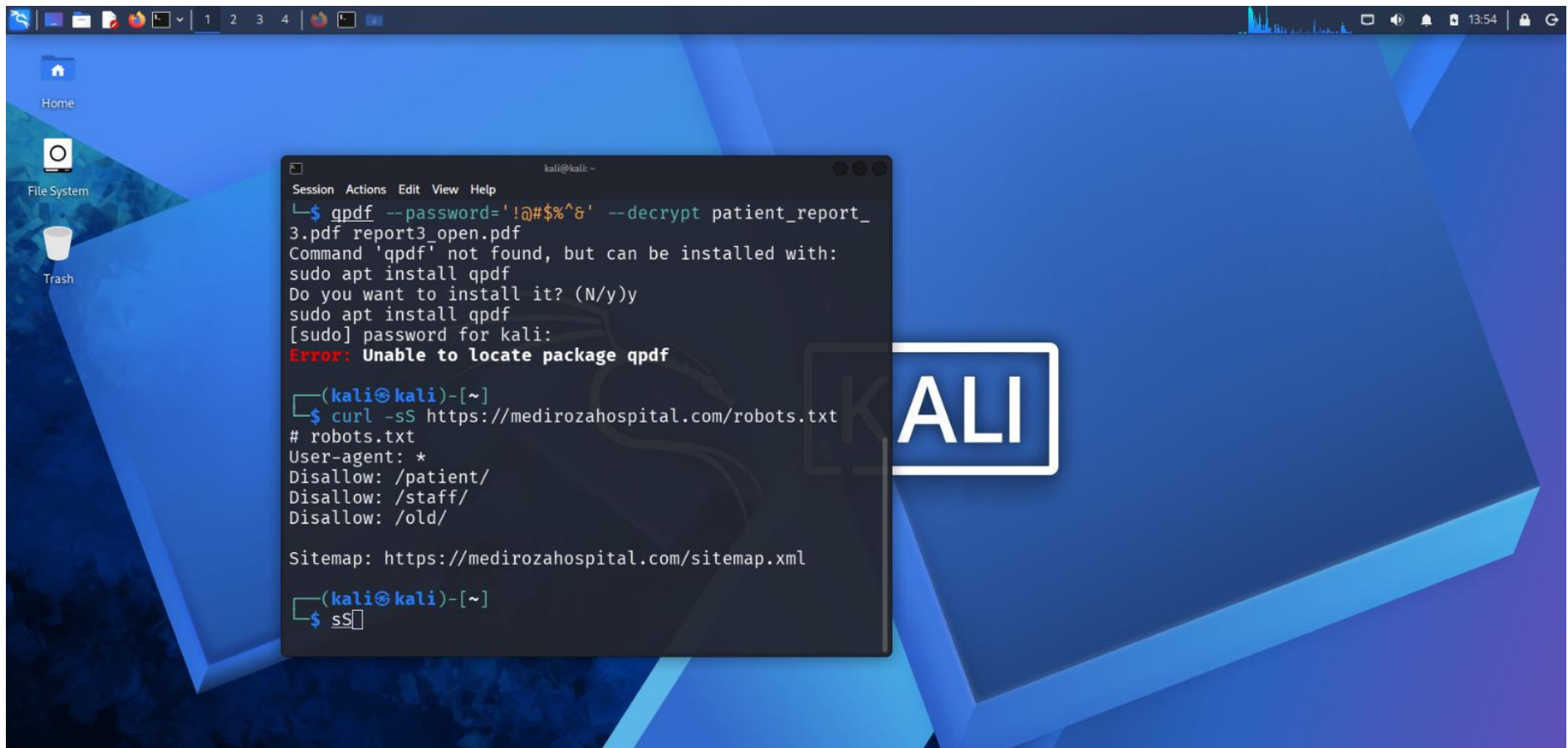
The tool reports a match for !@#\$%^& with 72,203 candidates tried from a 14,344,379-entry list. I used rockyou.txt for this attack; E09a shows the wordlist download page.



E10 M3 Backup discovery

robots.txt response and qpdf troubleshooting

curl returns Disallow entries including /old/. The preceding qpdf command and installation attempt failed; they are not evidence of successful decryption.



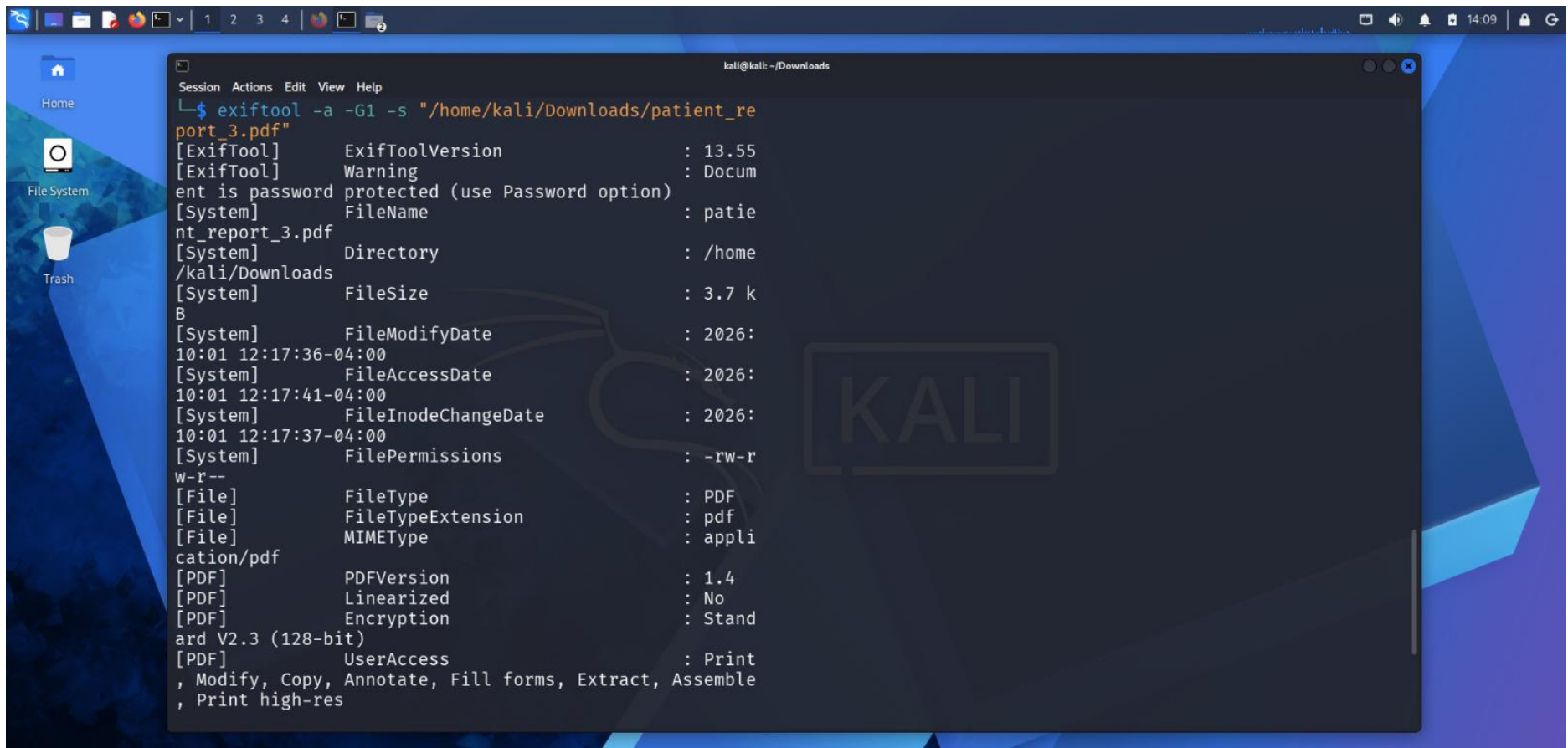
The screenshot shows a Kali Linux desktop with a blue geometric background. A terminal window is open in the center, displaying the following commands and output:

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ qpdf --password='!@#$$%^&' --decrypt patient_report_3.pdf report3_open.pdf  
Command 'qpdf' not found, but can be installed with:  
sudo apt install qpdf  
Do you want to install it? (N/y)y  
sudo apt install qpdf  
[sudo] password for kali:  
Error: Unable to locate package qpdf  
  
(kali@kali)-[~]  
└─$ curl -sS https://medirozahospital.com/robots.txt  
# robots.txt  
User-agent: *  
Disallow: /patient/  
Disallow: /staff/  
Disallow: /old/  
  
Sitemap: https://medirozahospital.com/sitemap.xml  
  
(kali@kali)-[~]  
└─$ ss
```

E11 M3 PDF metadata review

Initial ExifTool password warning

The first metadata read reports that the document is password protected and requests the Password option.



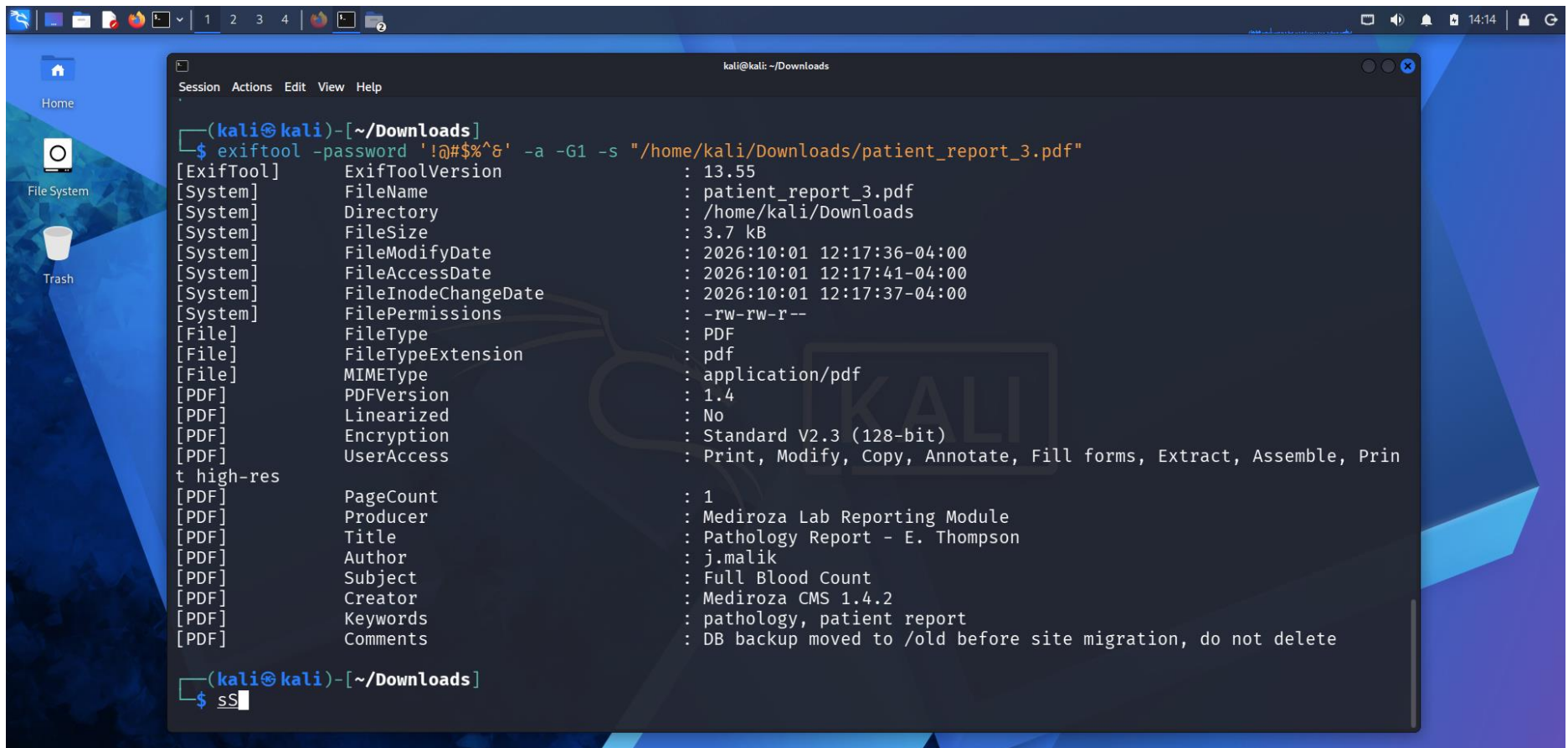
The screenshot shows a Kali Linux desktop environment. A terminal window is open, displaying the output of the command `exiftool -a -G1 -s "/home/kali/Downloads/patient_report_3.pdf"`. The output includes file metadata such as filename, directory, size, dates, permissions, and file type. A warning message indicates the document is password protected.

```
kali@kali: ~/Downloads
Session Actions Edit View Help
└─$ exiftool -a -G1 -s "/home/kali/Downloads/patient_report_3.pdf"
[ExifTool]      ExifToolVersion      : 13.55
[ExifTool]      Warning                : Document is password protected (use Password option)
[System]        FileName              : patient_report_3.pdf
[System]        Directory              : /home/kali/Downloads
[System]        FileSize               : 3.7 kB
[System]        FileModifyDate         : 2026:10:01 12:17:36-04:00
[System]        FileAccessDate         : 2026:10:01 12:17:41-04:00
[System]        FileInodeChangeDate    : 2026:10:01 12:17:37-04:00
[System]        FilePermissions        : -rw-rw-r--
[File]          FileType               : PDF
[File]          FileTypeExtension      : pdf
[File]          MIMEType               : application/pdf
[PDF]           PDFVersion             : 1.4
[PDF]           Linearized             : No
[PDF]           Encryption             : Standard V2.3 (128-bit)
[PDF]           UserAccess             : Print, Modify, Copy, Annotate, Fill forms, Extract, Assemble, Print high-res
```

E12 M3 PDF metadata review

Successful metadata read reveals the backup clue

ExifTool is run with the recovered password. The Comments field states that the database backup was moved to /old before site migration.



The screenshot shows a Kali Linux desktop environment. A terminal window is open, displaying the output of the ExifTool command. The command used is `exiftool -password '!@#$$%^&' -a -G1 -s "/home/kali/Downloads/patient_report_3.pdf"`. The output lists various file and PDF metadata fields. The 'Comments' field at the bottom of the output contains the text: 'DB backup moved to /old before site migration, do not delete'.

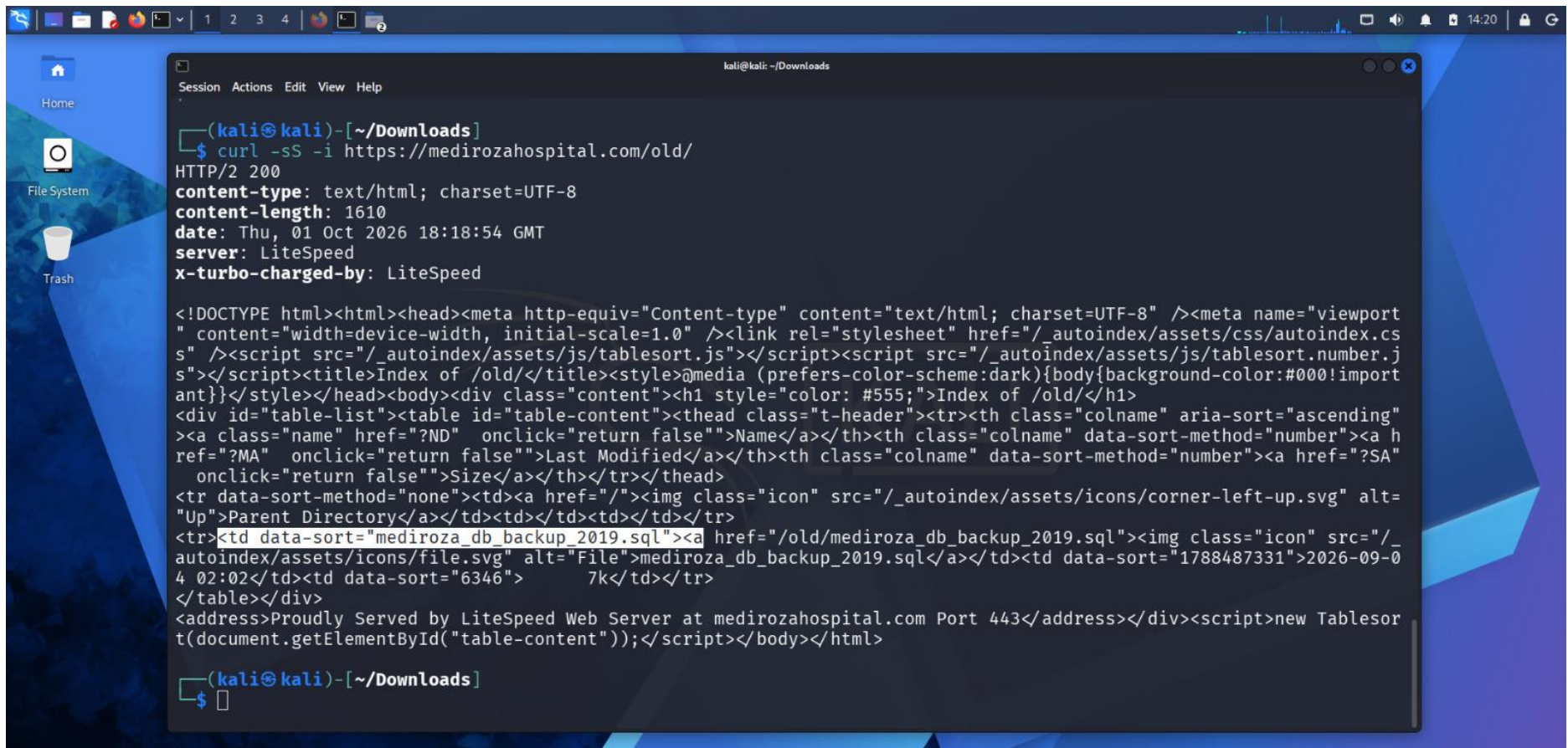
```
kali@kali: ~/Downloads
Session Actions Edit View Help
(kali@kali)-[~/Downloads]
$ exiftool -password '!@#$$%^&' -a -G1 -s "/home/kali/Downloads/patient_report_3.pdf"
[ExifTool]      ExifToolVersion      : 13.55
[System]        FileName              : patient_report_3.pdf
[System]        Directory            : /home/kali/Downloads
[System]        FileSize              : 3.7 kB
[System]        FileModifyDate       : 2026:10:01 12:17:36-04:00
[System]        FileAccessDate      : 2026:10:01 12:17:41-04:00
[System]        FileInodeChangeDate  : 2026:10:01 12:17:37-04:00
[System]        FilePermissions     : -rw-rw-r--
[File]          FileType              : PDF
[File]          FileTypeExtension    : pdf
[File]          MIMEType              : application/pdf
[PDF]           PDFVersion           : 1.4
[PDF]           Linearized           : No
[PDF]           Encryption           : Standard V2.3 (128-bit)
[PDF]           UserAccess            : Print, Modify, Copy, Annotate, Fill forms, Extract, Assemble, Prin
t high-res
[PDF]           PageCount            : 1
[PDF]           Producer            : Mediroza Lab Reporting Module
[PDF]           Title               : Pathology Report - E. Thompson
[PDF]           Author              : j.malik
[PDF]           Subject             : Full Blood Count
[PDF]           Creator             : Mediroza CMS 1.4.2
[PDF]           Keywords            : pathology, patient report
[PDF]           Comments            : DB backup moved to /old before site migration, do not delete

(kali@kali)-[~/Downloads]
$ ss
```

E13 M3 Backup discovery

HTTP 200 directory listing exposes the backup

The /old/ response contains an Index of /old/ page and a link to mediroza_db_backup_2019.sql.



```
kali@kali: ~/Downloads
Session Actions Edit View Help

(kali@kali)-[~/Downloads]
$ curl -sS -i https://medirozahospital.com/old/
HTTP/2 200
content-type: text/html; charset=UTF-8
content-length: 1610
date: Thu, 01 Oct 2026 18:18:54 GMT
server: LiteSpeed
x-turbo-charged-by: LiteSpeed

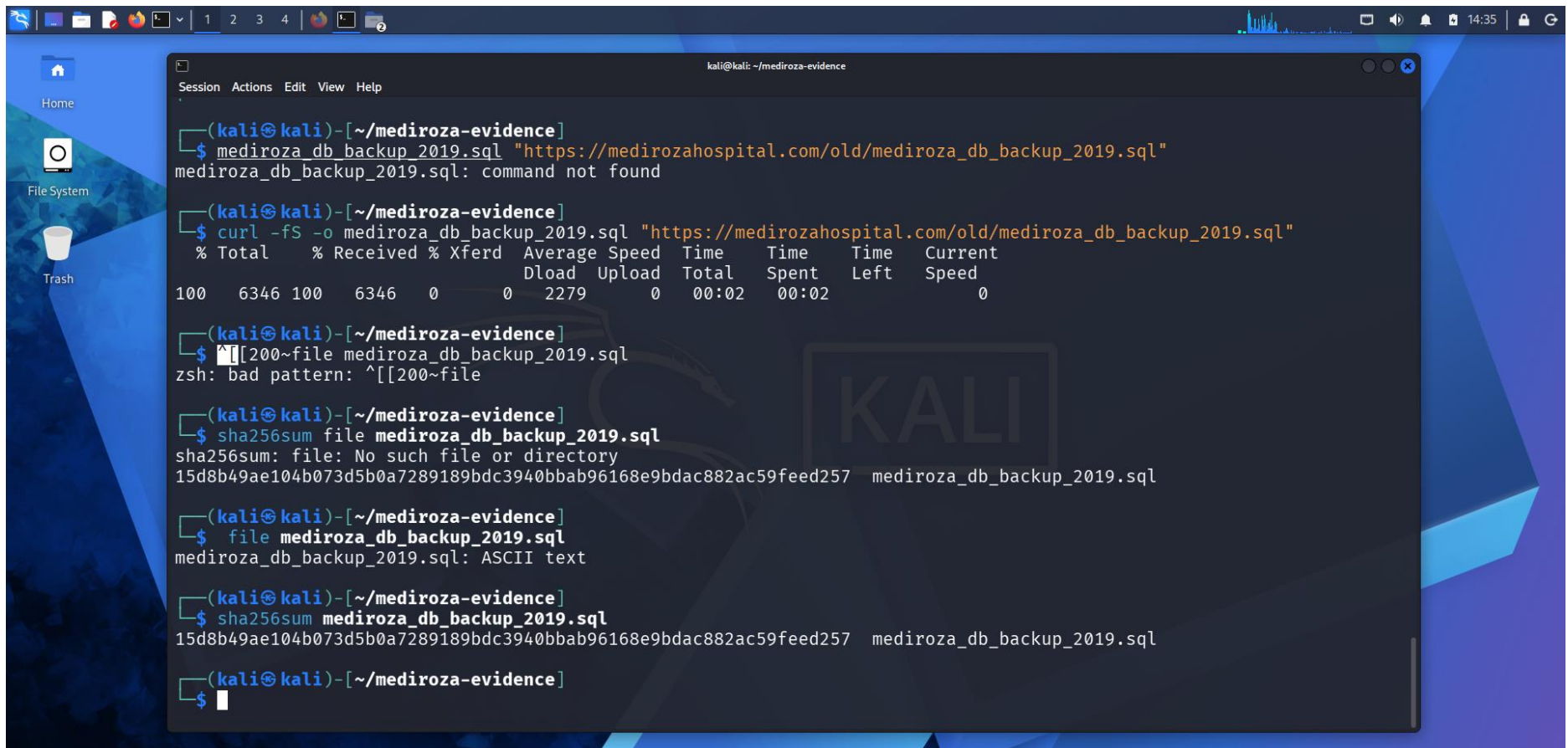
<!DOCTYPE html><html><head><meta http-equiv="Content-type" content="text/html; charset=UTF-8" /><meta name="viewport"
" content="width=device-width, initial-scale=1.0" /><link rel="stylesheet" href="/_autoindex/assets/css/autoindex.cs
s" /><script src="/_autoindex/assets/js/tablesort.js"></script><script src="/_autoindex/assets/js/tablesort.number.j
s"></script><title>Index of /old/</title><style>@media (prefers-color-scheme:dark){body{background-color:#000!import
ant}}</style></head><body><div class="content"><h1 style="color: #555;">Index of /old/</h1>
<div id="table-list"><table id="table-content"><thead class="t-header"><tr><th class="colname" aria-sort="ascending"
><a class="name" href="?ND" onclick="return false">Name</a></th><th class="colname" data-sort-method="number"><a href="?MA"
onclick="return false">Last Modified</a></th><th class="colname" data-sort-method="number"><a href="?SA"
onclick="return false">Size</a></th></tr></thead>
<tr data-sort-method="none"><td><a href="/">Parent Directory</a></td><td></td><td></td></tr>
<tr><td data-sort="mediroza_db_backup_2019.sql"><a href="/old/mediroza_db_backup_2019.sql">mediroza_db_backup_2019.sql</a></td><td data-sort="1788487331">2026-09-0
4 02:02</td><td data-sort="6346">7k</td></tr>
</table></div>
<address>Proudly Served by LiteSpeed Web Server at medirozahospital.com Port 443</address></div><script>new Tablesor
t(document.getElementById("table-content"));</script></body></html>

(kali@kali)-[~/Downloads]
$
```

E14 M3 Backup validation

Backup download and integrity checks

curl transfers 6,346 bytes. The final file command identifies ASCII text and the final sha256sum command produces a hash; earlier command errors remain visible.



```
kali@kali: ~/mediroza-evidence
Session Actions Edit View Help

(kali@kali)-[~/mediroza-evidence]
$ mediroza_db_backup_2019.sql "https://medirozahospital.com/old/mediroza_db_backup_2019.sql"
mediroza_db_backup_2019.sql: command not found

(kali@kali)-[~/mediroza-evidence]
$ curl -fs -o mediroza_db_backup_2019.sql "https://medirozahospital.com/old/mediroza_db_backup_2019.sql"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Dload  % Upload   Total   Spent    Left     Speed
100    6346  100    6346    0     0    2279      0  00:02  00:02   00:00    0

(kali@kali)-[~/mediroza-evidence]
$ [[200~file mediroza_db_backup_2019.sql
zsh: bad pattern: ^[[200~file

(kali@kali)-[~/mediroza-evidence]
$ sha256sum file mediroza_db_backup_2019.sql
sha256sum: file: No such file or directory
15d8b49ae104b073d5b0a7289189bdc3940bbab96168e9bdac882ac59feed257  mediroza_db_backup_2019.sql

(kali@kali)-[~/mediroza-evidence]
$ file mediroza_db_backup_2019.sql
mediroza_db_backup_2019.sql: ASCII text

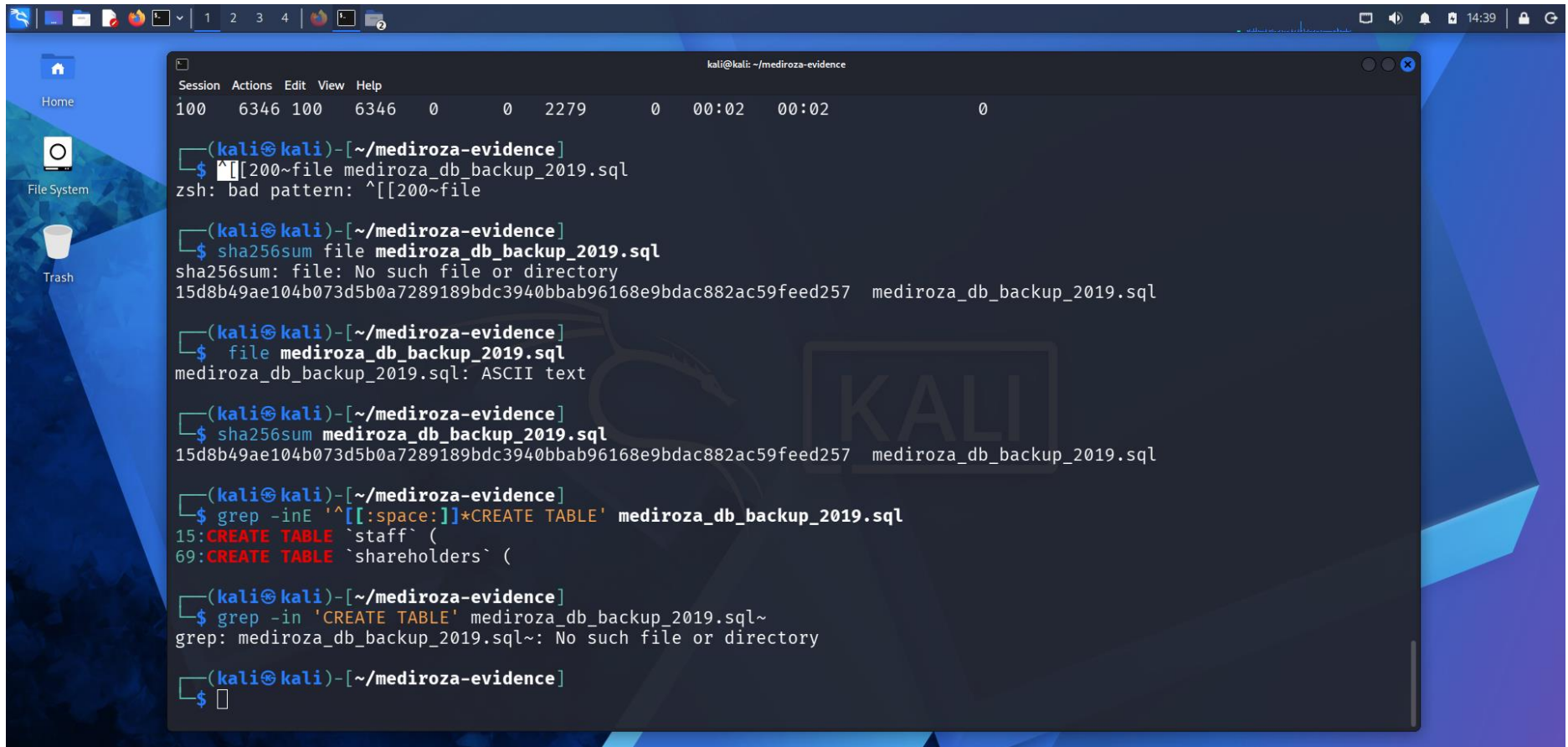
(kali@kali)-[~/mediroza-evidence]
$ sha256sum mediroza_db_backup_2019.sql
15d8b49ae104b073d5b0a7289189bdc3940bbab96168e9bdac882ac59feed257  mediroza_db_backup_2019.sql

(kali@kali)-[~/mediroza-evidence]
$
```

E15 M3 Table discovery

Staff and shareholders table locations

The successful grep output finds staff at line 15 and shareholders at line 69. A later command fails because the filename includes an extra tilde.



```
kali@kali: ~/mediroza-evidence
Session Actions Edit View Help
100 6346 100 6346 0 0 2279 0 00:02 00:02 0

(kali@kali)-[~/mediroza-evidence]
$ [[200~file mediroza_db_backup_2019.sql
zsh: bad pattern: ^[[200~file

(kali@kali)-[~/mediroza-evidence]
$ sha256sum file mediroza_db_backup_2019.sql
sha256sum: file: No such file or directory
15d8b49ae104b073d5b0a7289189bdc3940bbab96168e9bdac882ac59feed257 mediroza_db_backup_2019.sql

(kali@kali)-[~/mediroza-evidence]
$ file mediroza_db_backup_2019.sql
mediroza_db_backup_2019.sql: ASCII text

(kali@kali)-[~/mediroza-evidence]
$ sha256sum mediroza_db_backup_2019.sql
15d8b49ae104b073d5b0a7289189bdc3940bbab96168e9bdac882ac59feed257 mediroza_db_backup_2019.sql

(kali@kali)-[~/mediroza-evidence]
$ grep -inE '^[[[:space:]]*CREATE TABLE' mediroza_db_backup_2019.sql
15:CREATE TABLE `staff` (
69:CREATE TABLE `shareholders` (

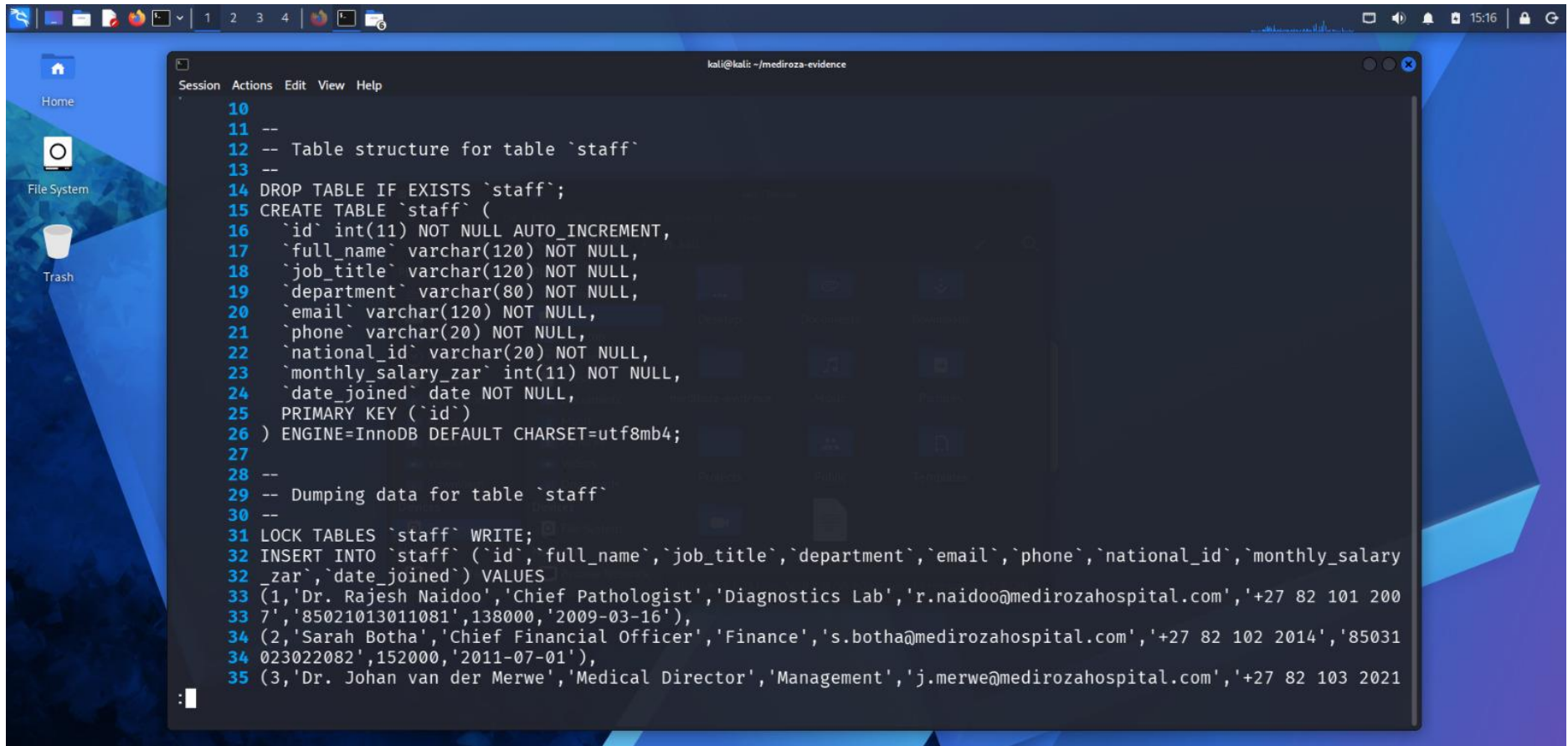
(kali@kali)-[~/mediroza-evidence]
$ grep -in 'CREATE TABLE' mediroza_db_backup_2019.sql~
grep: mediroza_db_backup_2019.sql~: No such file or directory

(kali@kali)-[~/mediroza-evidence]
$
```

E16 M3 Staff salary review

Staff schema and partial salary records

The view shows the staff schema, including monthly_salary_zar, and the beginning of its INSERT records. It is evidence of salary exposure, not a complete employee salary listing.



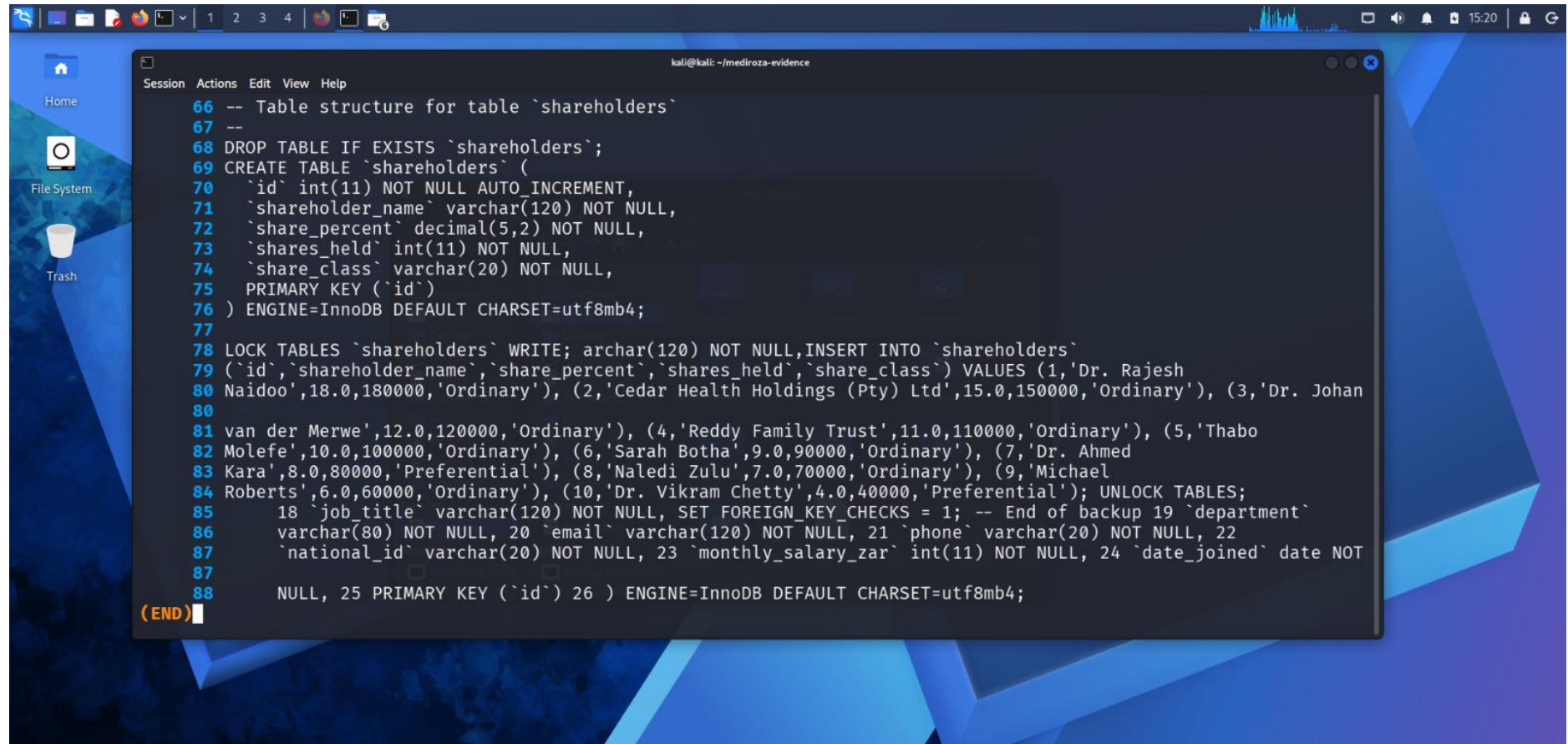
The screenshot shows a Kali Linux desktop environment. A terminal window is open, displaying a series of SQL commands. The commands are numbered 10 through 35. The terminal output shows the creation of a table named 'staff' with various columns including 'id', 'full_name', 'job_title', 'department', 'email', 'phone', 'national_id', 'monthly_salary_zar', and 'date_joined'. The table is created using InnoDB engine and utf8mb4 charset. Following the creation, the terminal shows the beginning of an INSERT statement, which is partially visible and appears to be dumping data for the 'staff' table. The data includes names like 'Dr. Rajesh Naidoo', 'Sarah Botha', and 'Dr. Johan van der Merwe', along with their job titles, departments, emails, phone numbers, national IDs, and monthly salaries in ZAR.

```
10 --
11 --
12 -- Table structure for table `staff`
13 --
14 DROP TABLE IF EXISTS `staff`;
15 CREATE TABLE `staff` (
16   `id` int(11) NOT NULL AUTO_INCREMENT,
17   `full_name` varchar(120) NOT NULL,
18   `job_title` varchar(120) NOT NULL,
19   `department` varchar(80) NOT NULL,
20   `email` varchar(120) NOT NULL,
21   `phone` varchar(20) NOT NULL,
22   `national_id` varchar(20) NOT NULL,
23   `monthly_salary_zar` int(11) NOT NULL,
24   `date_joined` date NOT NULL,
25   PRIMARY KEY (`id`)
26 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
27 --
28 --
29 -- Dumping data for table `staff`
30 --
31 LOCK TABLES `staff` WRITE;
32 INSERT INTO `staff` (`id`,`full_name`,`job_title`,`department`,`email`,`phone`,`national_id`,`monthly_salary
32 _zar`,`date_joined`) VALUES
33 (1,'Dr. Rajesh Naidoo','Chief Pathologist','Diagnostics Lab','r.naidoo@medirozahospital.com','+27 82 101 200
33 7','85021013011081',138000,'2009-03-16'),
34 (2,'Sarah Botha','Chief Financial Officer','Finance','s.botha@medirozahospital.com','+27 82 102 2014',85031
34 023022082',152000,'2011-07-01'),
35 (3,'Dr. Johan van der Merwe','Medical Director','Management','j.merwe@medirozahospital.com','+27 82 103 2021
```

E17 M3 Shareholder review

Shareholder schema and ownership records

The image shows shareholder_name, share_percent, shares_held and share_class, followed by records with identifiers 1 through 10. Wrapped and extraneous terminal text is preserved.

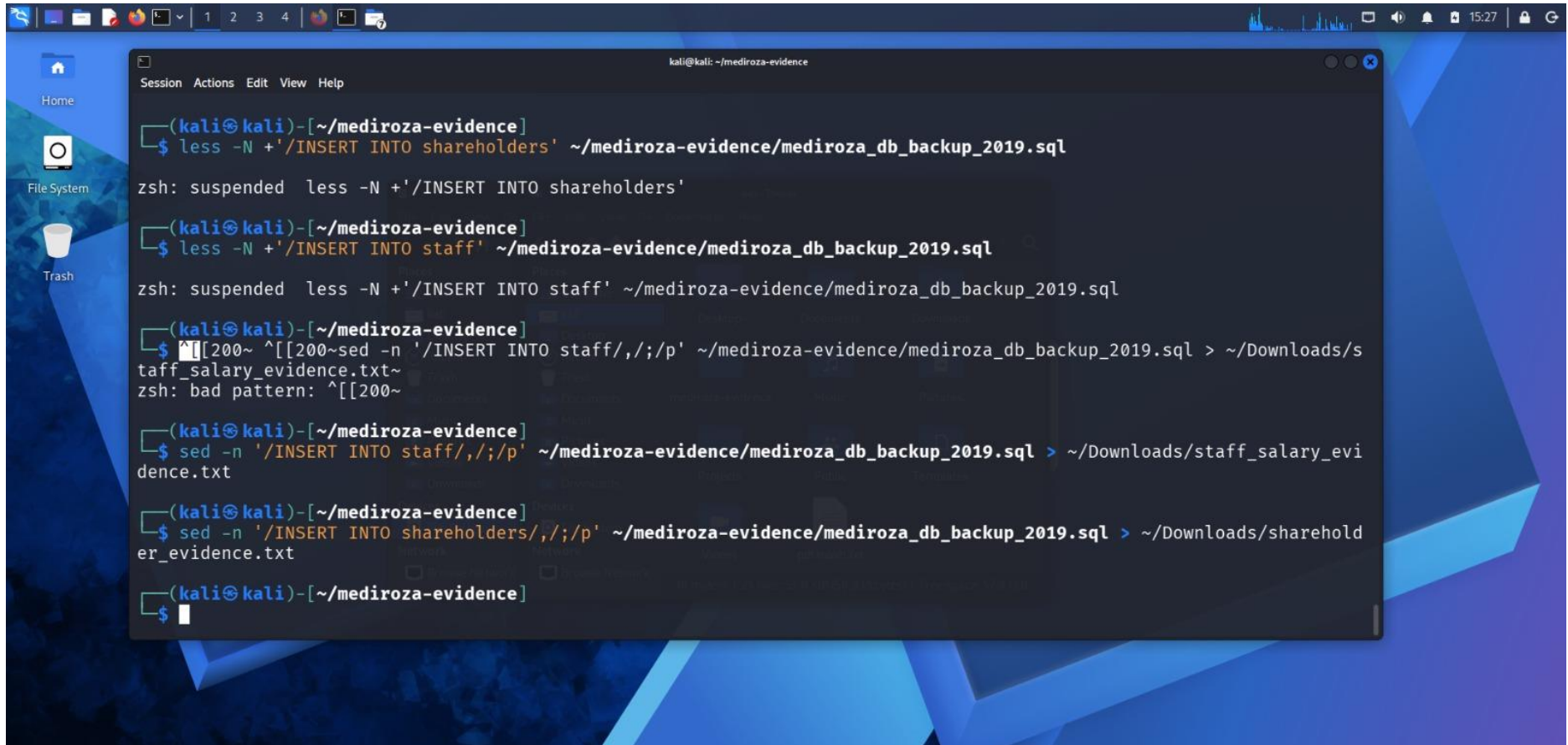


```
Session Actions Edit View Help
66 -- Table structure for table `shareholders`
67 --
68 DROP TABLE IF EXISTS `shareholders`;
69 CREATE TABLE `shareholders` (
70   `id` int(11) NOT NULL AUTO_INCREMENT,
71   `shareholder_name` varchar(120) NOT NULL,
72   `share_percent` decimal(5,2) NOT NULL,
73   `shares_held` int(11) NOT NULL,
74   `share_class` varchar(20) NOT NULL,
75   PRIMARY KEY (`id`)
76 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
77
78 LOCK TABLES `shareholders` WRITE; archar(120) NOT NULL,INSERT INTO `shareholders`
79 (`id`,`shareholder_name`,`share_percent`,`shares_held`,`share_class`) VALUES (1,'Dr. Rajesh
80 Naidoo',18.0,180000,'Ordinary'), (2,'Cedar Health Holdings (Pty) Ltd',15.0,150000,'Ordinary'), (3,'Dr. Johan
81 van der Merwe',12.0,120000,'Ordinary'), (4,'Reddy Family Trust',11.0,110000,'Ordinary'), (5,'Thabo
82 Molefe',10.0,100000,'Ordinary'), (6,'Sarah Botha',9.0,90000,'Ordinary'), (7,'Dr. Ahmed
83 Kara',8.0,80000,'Preferential'), (8,'Naledi Zulu',7.0,70000,'Ordinary'), (9,'Michael
84 Roberts',6.0,60000,'Ordinary'), (10,'Dr. Vikram Chetty',4.0,40000,'Preferential'); UNLOCK TABLES;
85   18 `job_title` varchar(120) NOT NULL, SET FOREIGN_KEY_CHECKS = 1; -- End of backup 19 `department`
86   varchar(80) NOT NULL, 20 `email` varchar(120) NOT NULL, 21 `phone` varchar(20) NOT NULL, 22
87   `national_id` varchar(20) NOT NULL, 23 `monthly_salary_zar` int(11) NOT NULL, 24 `date_joined` date NOT
88   NULL, 25 PRIMARY KEY (`id`) 26 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
(END)
```

E18 M3 Evidence export

Attempts to save staff and shareholder text files

The terminal shows less sessions and sed redirection commands. Exported contents are not displayed; the sed patterns omit the backticks around SQL table names, so file completeness is not established.



```
kali@kali: ~/mediroza-evidence
Session Actions Edit View Help

(kali@kali)-[~/mediroza-evidence]
$ less -N +'/INSERT INTO shareholders' ~/mediroza-evidence/mediroza_db_backup_2019.sql

zsh: suspended less -N +'/INSERT INTO shareholders'

(kali@kali)-[~/mediroza-evidence]
$ less -N +'/INSERT INTO staff' ~/mediroza-evidence/mediroza_db_backup_2019.sql

zsh: suspended less -N +'/INSERT INTO staff' ~/mediroza-evidence/mediroza_db_backup_2019.sql

(kali@kali)-[~/mediroza-evidence]
$ [[200~ ^[[200~sed -n '/INSERT INTO staff/,;/p' ~/mediroza-evidence/mediroza_db_backup_2019.sql > ~/Downloads/staff_salary_evidence.txt~
zsh: bad pattern: ^[[200~

(kali@kali)-[~/mediroza-evidence]
$ sed -n '/INSERT INTO staff/,;/p' ~/mediroza-evidence/mediroza_db_backup_2019.sql > ~/Downloads/staff_salary_evidence.txt

(kali@kali)-[~/mediroza-evidence]
$ sed -n '/INSERT INTO shareholders/,;/p' ~/mediroza-evidence/mediroza_db_backup_2019.sql > ~/Downloads/shareholder_evidence.txt

(kali@kali)-[~/mediroza-evidence]
$
```